WELLSLING · RISK REGISTER

WellSling risk register and hazard analysis

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	First issue (kit 1.9.1 upgrade) with the project risks, the preliminary hazard analysis and a first-pass design FMEA of the chosen design (option A)

WellSling risk register and hazard analysis

The top risks are a person going down the well despite the kit, a falling or swinging 600 kg load, and harm to the animal from the bands or the time it hangs. The design answers the first two with a method that needs nobody below the rim, an automatic load brake, factors of 3 or more and a proof-load gate; the third needs a veterinary partner. Method: STANDARDS section 23. Likelihood, impact, severity, occurrence and detection are scored 1 (low) to 5 (high); for detection, 5 means hard to detect.

> **Safety:** Nobody goes down the well at any stage; a person in a well is a call to the emergency services. The kit is an open engineering reference, not certified lifting or rescue equipment, and is not used on an animal before a proof-load test at 1.5 times the hook load.

Project risk register

Table 1. Project risks.

#	Risk	Cause	Effect	Likelihood	Impact	Owner (role)	Response
P1	No co-design partner takes it up	Partners not yet approached; well rescues are sporadic	No trial wells, animals or veterinary review; stalls at TRL 3	3	4	Amish	Approach the named candidates in WSL-PRB-001 with the build plan and renders
P2	Veterinary review finds the bands or lift time harmful	No cited tolerance for cattle in slings	Band width, padding or lift speed must change	3	3	Veterinary partner	Review before any live lift; wider bands or a third band are options
P3	Winch with a load-pressure brake not available locally	Spur-gear braked winches are less common than worm winches	Cost or lead time; a worm winch would make the lift too slow	2	3	Design lead	Check suppliers in the partner's region; keep the brake requirement (PR-007)
P4	Cost stays about twice the value-engineering target	Own tripod, winch and bridge	Harder for a village to buy	4	2	Amish	Shared kits per cluster of villages; savings in WSL-DEC-001 value engineering
P5	Passer cannot pass the bands under a lying animal	Long flexible pole, mud, a wedged animal	The method stalls at the first step	3	4	Design lead	Dummy trial in a test pit at TRL 4 (STEM S26); stiffer lower sections if needed
P6	Users improvise without the proof-load test or the safety stops	Urgency of a real rescue	Collapse or a dropped animal	3	5	Safety lead	Proof-load gate and safety stops in WSL-BLD-001; training with the partner

Preliminary hazard analysis

Table 2. Hazards.

#	Hazard	Cause	Harm	Severity	Likelihood	Control	Residual
H1	Bad air in the well	A person climbs down to help	Death of rescuers, often several	5	2	Method needs nobody below the rim (PR-001); safety notes in every document; call the emergency services for a person in a well	Low, depends on discipline at the well
H2	Falling load	Rope, shackle, pin or band fails; brake slips	Animal killed or injured; people struck	5	2	Factors of 3 or more, rope 6.0; automatic load brake; proof load 1.5 times before use; nobody under the load	Low after the proof-load test
H3	Tripod tips or a foot slides	Guide line pulled hard, swing, soft ground	Frame falls across the well and the crew	5	2	Tips only at 19.4 deg (PR-010); spread chains; staked pads; hook line kept within 10 deg	Low
H4	Rim collapses	Crumbling or undercut rim loaded by people or pads	Person or frame falls into the well	5	2	Pads 700 mm and bearers 400 mm back; rim and lining checked at safety stop S4; harnesses within 1 m (PR-014)	Low to medium
H5	Person falls into the well	Working at the rim, reaching for lines	Injury or death	5	2	Harness on a work-restraint lanyard tied to a foot	Low
H6	Animal injured by the bands	Pressure and time in the bands; uneven share; regurgitation	Injury or death of the animal	4	3	Two 240 mm padded bands; level the animal with the chains; veterinary review before live lifts	Medium until the review
H7	Kicking or thrashing animal	Fright as it is lifted	Struck crew; dynamic load on the rig	4	3	Hands clear of bands and legs; guide lines used gently; the 1.5 times proof covers a dynamic factor of 1.5 (STEM S36); vet or handler present	Medium
H11	Animal stuck in mud	Suction and adhesion at the well bottom	Overload of the rig if the crank is forced	4	3	Stop if the crank gets much harder with no movement (safety stop S5); breakout force to be measured in the dummy trial (STEM S39)	Medium
H8	Crank handle kickback	Handle released without the brake engaging	Hand or arm struck	3	1	Load-pressure brake holds whenever the handle is released; brake test at proof load	Low
H9	Pinch points			3	3		Low

#	Hazard	Cause	Harm	Severity	Likelihood	Control	Residual
		Pins, shackles, chain links, bridge runners pushed under the load	Crushed fingers			Runners pushed from the side, never reached under the animal; gloves	
H10	Struck by a falling part during setup	Walking the tripod up; head or sheave dropped	Head or foot injury	4	2	Head pinned on the ground; two foot ropes; nobody under the head; no climbing	Low

Design FMEA (first pass, P3)

Table 3. Failure modes of option A. RPN = severity x occurrence x detection.

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
F1	Leg 1 (winch leg)	Carry compression and the rope's bending	Buckles	Frame collapses	Bent leg, low-grade tube, overload	5	1	3	15	88.9 x 3.2 mm structural tube, factor 3.7 with an initial bow (D10); straightness check in the plan; mill certificate
F2	Leg pins and R-clips	Hold the legs to head and feet	Pin walks out	Leg drops free	R-clip missing or not on its lanyard	5	2	2	20	R-clips on lanyards; check at every hold point
F3	Spread chains and fins	Stop the feet sliding out	Chain or shackle comes off	Feet slide; tripod collapses	Unmoused pin; slack chain	5	2	2	20	One fin and shackle per chain, in line (D11); mouse every pin; take up slack
F4	Winch brake	Hold the load when the handle is released	Brake slips	Load runs back	Wrong winch type; worn brake; oil on the friction discs	5	2	3	30	Load-pressure brake specified; brake hold test at proof load; inspect before use
F5	Winch bolts	Hold the winch on the bracket	Bolt loosens	Winch tilts; rope fleet changes	Nyloc not replaced	4	2	2	16	All four bolts clear the tube (D12); nyloc nuts; torque check
F6	Wire rope	Carry the hook load	Breaks	Load falls	Kinks, broken wires, crushing on the drum	5	1	2	10	Factor 6.0; D/d 25; inspect before each use; replace on damage
F7	Sheave and axle	Turn the rope over the head	Axle comes out	Rope drops off the head	Retaining ring missing	5	1	3	15	Head on the far side, DIN 471 ring on the leg side (D10); check at safety stop
F8	Lift shackle	Join hook to spreader	Pin unscrews	Spreader drops	Unmoused screw pin	5	2	2	20	Mouse with wire; latched hook
F9	Band end yokes	Carry the band eye on a straight bar	Removable bar slides out	Band end drops	R-clip missing	5	2	2	20	Two R-clips per bar; check before lifting (D13)
F10	Webbing bands	Carry and cradle the animal	Band cut or torn	Animal slips or falls	Sharp edge, alkali damage, UV	5	2	2	20	Felt sleeves; inspect; store dry and dark; no lime or ash (STEM S29)
F11			Uneven share; slips			4	3	3	36	Choose sleeves 300 to 500 mm; adjust

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
	Bands on the animal	Spread the load	toward the belly	Injury; animal tips	Band spacing wrong for the animal					chains to level; veterinary guidance (S25)
F12	Pads and stakes	Carry the feet on soft ground	Pad sinks or tilts	Tripod leans	Soft or saturated ground	4	2	2	16	15 kPa design pressure; larger pads on soft ground; site check
F13	Bridge runners	Carry the animal on landing	Runner slips off a bearer	Deck drops	Runner not over the bearer band	4	2	2	16	White bands mark the bearing length; cleats keep the deck on
F14	Deck panels	Carry the hooves	Plywood breaks under a hoof	Leg through the deck	Rotten or delaminated ply	4	1	2	8	Factor 3.4; exterior ply; sealed edges; inspect
F15	Passer pole joints	Keep the pole together	Section separates down the well	Pole lost; J-head down the well	Button lock not engaged	2	3	2	12	Button must show in every joint; tether line on the J-head